

随机变量的数字特征

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1. 由题可知:

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{2\pi}e^{-\frac{x^2+y^2}{2}} + \frac{1}{2\pi}e^{-\pi^2}\cos x \cos y, & -\pi < x, y < \pi \\ \frac{1}{2\pi}e^{-\frac{x^2+y^2}{2}}, & \text{else} \end{cases}$$

首先验证非负性, 当 $x \notin (-\pi, \pi)$ 或 $y \notin (-\pi, \pi)$ 时, 显然有 $f_{X,Y}(x,y) > 0$.

当 $x, y \in (-\pi, \pi)$ 时,

$$f_{X,Y}(x,y) > \frac{1}{2\pi}e^{-\frac{\pi^2+\pi^2}{2}} - \frac{1}{2\pi}e^{-\pi^2} = 0$$

故 $f_{X,Y} \geq 0$.

下面验证 $\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f_{X,Y}(x,y) dx dy = 1$.

$$\begin{aligned} & \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f_{X,Y}(x,y) dx dy \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} e^{-\frac{x^2+y^2}{2}} dx dy + \frac{e^{-\pi^2}}{2\pi} \int_{-\pi}^{+\pi} \int_{-\pi}^{+\pi} \cos x \cos y dx dy \\ &= 1 + 0 \\ &= 1 \end{aligned}$$

故 $f_{X,Y}$ 是概率密度函数.

(1) 由边缘概率密度的定义:

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{+\infty} f_{X,Y}(x,y) dy \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} e^{-\frac{x^2+y^2}{2}} dy + \frac{e^{-\pi^2}}{2\pi} \int_{-\pi}^{+\pi} \cos x \cos y dy \\ &= \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{y^2}{2}} dy \\ &= \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \end{aligned}$$

故 $X \sim N(0, 1)$. 根据对称性同理可得 $Y \sim N(0, 1)$. □

(2) 由 (1) 可知: $\exists x, y, f_{X,Y}(x,y) \neq f_X(x)f_Y(y)$, 故 X 和 Y 不独立.

下证 $\rho_{XY} = 0$, 只需证 $\mathbb{E}(XY) = \mathbb{E}(X)\mathbb{E}(Y) = 0$.

$$\begin{aligned}\mathbb{E}(XY) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy f_{X,Y}(x,y) dx dy \\ &= \frac{1}{2\pi} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} xy \cdot e^{-\frac{x^2+y^2}{2}} dx dy + \frac{e^{-\pi^2}}{2\pi} \int_{-\pi}^{+\pi} \int_{-\pi}^{+\pi} xy \cos x \cos y dx dy \\ &= 0 + 0 \\ &= 0\end{aligned}$$

□

2. (1) 根据期望的性质有:

$$\begin{aligned}1 &= \mathbb{E}(1) \\ &= \mathbb{E}\left(\frac{\sum_{i=1}^n X_i}{\sum_{i=1}^n X_i}\right) \\ &= \sum_{i=1}^n \mathbb{E}\left(\frac{X_i}{\sum_{i=1}^n X_i}\right) \\ &= n\mathbb{E}\left(\frac{X_1}{\sum_{i=1}^n X_i}\right)\end{aligned}$$

因此, $\mathbb{E}\left(\frac{X_1}{\sum_{i=1}^n X_i}\right) = \frac{1}{n}$.

(2) 由特征函数的定义知:

$$\begin{aligned}\psi_X(t) &= \mathbb{E}(e^{itX}) \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{itx} e^{-\frac{x^2}{2}} dx \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-\frac{1}{2}(x-it)^2 - \frac{t^2}{2}} dx \\ &= e^{-\frac{t^2}{2}}\end{aligned}$$

注意到当 $k \in \mathbb{N}_+$ 时

$$\begin{aligned}\psi_X^{(2k-1)}(0) &= 0 \\ \psi_X^{(2k)}(0) &= -(2k-1)\psi_X^{(2k-2)}(0)\end{aligned}$$

又由 $\psi_X^{(0)}(0) = 1$ 可得

$$\begin{aligned}\mathbb{E}(X^n) &= \begin{cases} 1, & n = 0 \\ 0, & n \in \{2k-1 \mid k \in \mathbb{N}_+\} \\ (-i)^n (-1)^{\frac{n}{2}} \prod_{i=1}^{\frac{n}{2}} (2i-1), & n \in \{2k \mid k \in \mathbb{N}_+\} \end{cases} \\ &= \begin{cases} 1, & n = 0 \\ 0, & n \in \{2k-1 \mid k \in \mathbb{N}_+\} \\ \prod_{i=1}^{\frac{n}{2}} (2i-1), & n \in \{2k \mid k \in \mathbb{N}_+\} \end{cases}\end{aligned}$$

(3) 令 $Z = \min\{X, Y\}$ 首先计算 $f_Z(z)$.

$$\begin{aligned} F_Z(z) &= P(Z \leq z) \\ &= P(\min\{X, Y\} \leq z) \\ &= 1 - P(X > z, Y > z) \\ &= 1 - P(X > z)P(Y > z) \\ &= 1 - (1 - F_X(z))(1 - F_Y(z)) \end{aligned}$$

由题可知:

$$F_X(t) = 1 - e^{-xt}$$

$$F_Y(t) = 1 - e^{-yt}$$

故

$$\begin{aligned} F_Z(z) &= 1 - (1 - F_X(z))(1 - F_Y(z)) \\ &= 1 - e^{-(x+y)z} \\ f_Z(z) &= \frac{dF_Z(z)}{dz} \\ &= (x+y)e^{-(x+y)z} \end{aligned}$$

即 $Z \sim \text{Exp}(x+y)$, 故 $\mathbb{E}(\min\{X, Y\}) = \mathbb{E}(Z) = \frac{1}{x+y}$.

3. (1) 根据矩生成函数的定义可知

$$\begin{aligned} f_X(t) &= \mathbb{E}(e^{tX}) \\ &= \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^{+\infty} e^{-\left(\frac{x}{\sqrt{2}\sigma} - \frac{\sigma t}{\sqrt{2}}\right)^2 + \frac{\sigma^2 t^2}{2}} dx \\ &= e^{\sigma^2 t^2} \end{aligned}$$

故 X 是 σ -subgaussian 的. □

(2) 根据矩生成函数的定义可知

$$\begin{aligned} f_X(t) &= \mathbb{E}(e^{tX}) \\ &= \lambda \int_0^{+\infty} e^{(t-\lambda)x} dx \end{aligned}$$

注意到该无穷积分在 $t \geq \lambda$ 时发散到正无穷, 故对于 $\forall \sigma, \exists \epsilon_0 > 0, t_0 \in (\lambda - \epsilon_0, \lambda)$, 使得

$$f_X(t_0) = \frac{\lambda}{\lambda - t_0} > \frac{\lambda}{\epsilon_0} > e^{\frac{\sigma^2 t_0^2}{2}}$$

. 即对于 $\forall \sigma$, X 都不是 σ -subgaussian 的. □

4. (1) 由题可知: X_k 表示从第一次拥有 $k-1$ 种卡到第一次拥有 k 种卡的抽卡次数. 由于每次抽卡均为伯努利试验, 因此在拥有 $k-1$ 种卡的条件下每次抽卡抽到新卡的概率 $p_k = \frac{n-k+1}{n}$. 故 $X_k \sim G(p_k)$.

由于每次独立均匀抽卡，故先前抽到新卡所需的抽卡次数与当前在已经有 $k-1$ 种卡的条件下抽到新卡所需的抽卡次数无关，即

$$\begin{aligned} & P(X_1 \leq x_1, X_2 \leq x_2, \dots, X_n \leq x_n) \\ &= P(X_1 \leq x_1)P(X_2 \leq x_2 \mid X_1 \leq x_1)P(X_3 \leq x_3 \mid X_1 \leq x_1, X_2 \leq x_2) \dots \\ &= P(X_1 \leq x_1)P(X_2 \leq x_2)P(X_3 \leq x_3) \dots \end{aligned}$$

故 X_k 相互独立. □

(2) 根据期望的性质可得：

$$\begin{aligned} \mathbb{E}(T_n) &= \mathbb{E}(X_1 + X_2 + \dots + X_n) \\ &= \sum_{k=1}^n \frac{n}{n-k+1} \\ &= nH_n \end{aligned}$$

(3) 利用特征函数进行证明. 注意到 $\mathbb{E}(e^{-\frac{T_n}{n}}) = \psi_{T_n}(\frac{i}{n})$ ，根据特征函数的性质可得：

$$\begin{aligned} \mathbb{E}(e^{-\frac{T_n}{n}}) &= \prod_{k=1}^n \psi_{X_k}(\frac{i}{n}) \\ &= \prod_{k=1}^n \frac{n-k+1}{e^{\frac{1}{n}n-k+1}} \\ &\leq \prod_{k=1}^n \frac{n-k+1}{(\frac{1}{n}+1)n-k+1} \\ &= \prod_{k=1}^n \frac{n-k+1}{n-k+2} \\ &= \prod_{k=1}^n \frac{k}{k+1} \\ &= \frac{1}{n+1} \end{aligned}$$

□

5. (1) 设事件 $A_{ij} = \{i, j\text{-之间有边}\}, i < j$ ，则

$$1_{A_{ij}} = \begin{cases} 1, & A_{ij} \text{发生} \\ 0, & A_{ij} \text{不发生} \end{cases}$$

因此有

$$\begin{aligned}
 X_0 &= \sum_{i=-\infty}^0 \sum_{j=1}^{+\infty} 1_{A_{ij}} \\
 \mathbb{E}(X_0) &= \sum_{i=-\infty}^0 \sum_{j=1}^{+\infty} \frac{1}{(j-i)^3} \\
 &= \sum_{j=1}^{+\infty} \sum_{i=0}^{+\infty} \frac{1}{(j+i)^3} \\
 &= \zeta(3) + (\zeta(3) - \frac{1}{1^3}) + (\zeta(3) - \frac{1}{1^3} - \frac{1}{2^3}) + \dots \\
 &= \zeta(3) + \sum_{n=1}^{+\infty} \left(\zeta(3) - \sum_{m=1}^n \frac{1}{m^3} \right)
 \end{aligned}$$

(2) 由 (1) 同理可得:

$$X_1 = \sum_{i=-\infty}^1 \sum_{j=2}^{+\infty} 1_{A_{ij}}$$

注意到

$$\begin{aligned}
 X_1 &= \sum_{i=-\infty}^0 \sum_{j=2}^{+\infty} 1_{A_{ij}} + \sum_{j=2}^{+\infty} 1_{A_{1j}} \\
 X_0 &= \sum_{i=-\infty}^0 \sum_{j=2}^{+\infty} 1_{A_{ij}} + \sum_{i=-\infty}^0 1_{A_{i1}}
 \end{aligned}$$

故

$$\begin{aligned}
 \text{Cov}(X_0, X_1) &= \text{Cov}\left(\sum_{i=-\infty}^0 \sum_{j=2}^{+\infty} 1_{A_{ij}}, \sum_{i=-\infty}^0 \sum_{j=2}^{+\infty} 1_{A_{ij}}\right) \\
 &\quad + \text{Cov}\left(\sum_{i=-\infty}^0 \sum_{j=2}^{+\infty} 1_{A_{ij}}, \sum_{i=-\infty}^0 1_{A_{i1}}\right) \\
 &\quad + \text{Cov}\left(\sum_{i=-\infty}^0 \sum_{j=2}^{+\infty} 1_{A_{ij}}, \sum_{j=2}^{+\infty} 1_{A_{1j}}\right) \\
 &\quad + \text{Cov}\left(\sum_{i=-\infty}^0 1_{A_{i1}}, \sum_{j=2}^{+\infty} 1_{A_{1j}}\right) \\
 &= \sum_{i=-\infty}^0 \sum_{j=2}^{+\infty} \text{Var}(1_{A_{ij}}) \\
 &= \sum_{i=-\infty}^0 \sum_{j=2}^{+\infty} \left(\frac{1}{(j-i)^3} - \frac{1}{(j-i)^6} \right)
 \end{aligned}$$

6. 由题可知:

$$\begin{aligned}
 \mathbb{E}(X) &= \mathbb{E}(Y) = 0 \\
 \text{Var}(X) &= \mathbb{E}(X^2) = 1 \\
 \text{Var}(Y) &= \mathbb{E}(Y^2) = 1 \\
 \text{Cov}(X, Y) &= \mathbb{E}(XY) = \rho
 \end{aligned}$$

(1) 由题可知:

$$\mathbb{E}((X+Y)^2) = \mathbb{E}(X^2) + \mathbb{E}(Y^2) + 2\mathbb{E}(XY) = 2 + 2\rho$$

$$\mathbb{E}((X-Y)^2) = \mathbb{E}(X^2) + \mathbb{E}(Y^2) - 2\mathbb{E}(XY) = 2 - 2\rho$$

(2) 由题可得:

$$\begin{aligned} \mathbb{E}(\max\{X^2, Y^2\}) &= \frac{1}{2}\mathbb{E}(X^2) + \frac{1}{2}\mathbb{E}(Y^2) + \frac{1}{2}\mathbb{E}(|X^2 - Y^2|) \\ &= 1 + \frac{1}{2}\mathbb{E}(|X^2 - Y^2|) \\ &\leq 1 + \frac{1}{2}\sqrt{\mathbb{E}((X+Y)^2)\mathbb{E}((X-Y)^2)} \\ &= 1 + \sqrt{1 - \rho^2} \end{aligned}$$

□

7. 本题为抽签问题, 对于 $\forall k \in \{1, 2, \dots, n\}, X_k \in \{0, 1\}$.

$$P(X_k = 1) = \frac{C_{N-1}^{a-1}}{C_N^a} = \frac{a}{N} = p$$

$$P(X_k = 0) = q$$

$$X = \sum_{k=1}^n X_k.$$

$$(1) \quad \mathbb{E}(X) = \sum_{i=1}^n \mathbb{E}(X_i) = np.$$

$$(2) \quad \text{Var}(X) = \text{Var}(a) = 0.$$

$$\text{Cov}(X_i, X_j) = \mathbb{E}(X_i X_j) - \mathbb{E}(X_i)\mathbb{E}(X_j) = \frac{a-1}{N-1}p - p^2, 1 \leq i < j \leq n.$$

(3)

$$\begin{aligned} \text{Var}(X) &= \text{Var}\left(\sum_{i=1}^n X_i\right) \\ &= \sum_{i=1}^n \text{Var}(X_i) + \sum_{i \neq j} \text{Cov}(X_i, X_j) \\ &= npq + n(n-1) \left(\frac{a-1}{N-1}p - p^2 \right) \\ &= np \left(q + (n-1) \left(\frac{a-1}{N-1} - p \right) \right) \end{aligned}$$

8. 设无向图 $G = (V, E), V = \{v_1, v_2, \dots, v_n\}$. 设事件 $A_{ijk} = \{(v_i, v_j), (v_j, v_k), (v_k, v_i) \in E \mid 1 \leq i < j < k \leq n\}$.

(1) 注意到 $X = \sum_{i < j < k} 1_{A_{ijk}}$, 故

$$\mathbb{E}(X) = C_n^3 P(A_{ijk}) = \frac{n(n-1)(n-2)}{6} p^3$$

(2) 根据方差的性质有

$$\text{Var}(X) = \sum_{i < j < k} \text{Var}(1_{A_{ijk}}) + \sum_{(i_1, j_1, k_1) \neq (i_2, j_2, k_2), i_* < j_* < k_*} \text{Cov}(1_{A_{i_1 j_1 k_1}}, 1_{A_{i_2 j_2 k_2}})$$

注意到只有当 (i_1, j_1, k_1) 和 (i_2, j_2, k_2) 中有且仅有两个相同时有

$$\text{Cov}(1_{A_{i_1 j_1 k_1}}, 1_{A_{i_2 j_2 k_2}}) \neq 0$$

因此有

$$\begin{aligned} \text{Var}(X) &= \sum_{i < j < k} \text{Var}(1_{A_{ijk}}) + C_n^2(n-2)(n-3)((p \cdot p^2 \cdot p^2) - (p^3)^2) \\ &= \frac{n(n-1)}{2} \left(-\frac{n+1}{3}p^6 + p^5 + \frac{n-2}{3}p^3 \right) \end{aligned}$$

9. (1) 假设存在可逆矩阵 A , 使得 $\mathbf{Z} = A\mathbf{Y} \sim N(0, \Sigma)$.

根据密度变换公式:

$$\begin{aligned} f_{\mathbf{Z}}(\mathbf{z}) &= f_{\mathbf{Y}}(\mathbf{y}) \left| \det \frac{d\mathbf{y}}{d\mathbf{z}} \right| \\ &= \frac{1}{(2\pi)^{\frac{n}{2}}} e^{-\frac{1}{2}\mathbf{y}^T \mathbf{y}} |A^{-1}| \\ &= \frac{1}{(2\pi)^{\frac{n}{2}} |A|} e^{-\frac{1}{2}(A^{-1}\mathbf{z})^T (A^{-1}\mathbf{z})} \\ &= \frac{1}{(2\pi)^{\frac{n}{2}} |A|} e^{-\frac{1}{2}\mathbf{z}^T (AA^T)^{-1} \mathbf{z}} \end{aligned}$$

由 $\mathbf{Z} \sim N(0, \Sigma)$ 知:

$$AA^T = \Sigma$$

由于 Σ 为实对称半正定矩阵, 因此必然存在特征矩阵 V 以及对应的特征值对角矩阵 D , 其中 $D_{ii} \geq 0, i = 1, 2, \dots, n$, 使得

$$\Sigma = VDV^T = (VD^{\frac{1}{2}})(VD^{\frac{1}{2}})^T$$

因此, 存在 $A = VD^{\frac{1}{2}}$ 使得 $\mathbf{Z} \sim N(0, \Sigma)$. □